Sam Schwarzkopf
Wednesday, 28 September 2026

Delineating visual ROIs with SamSrf

As of *SamSrf 10.4* we have retired the old tools that could be used to delineate visual areas directly in MatLab, specifically *DelineationTool* and *AutoDelineation*. They are still available and located in the *SamSrf/Utils* folder; however, recent Matlab updates have effectively ruined the interface and there are several other issues. Therefore, we no longer support them and I'm afraid if you run into issues with them, you're on your own. We have replaced this documentation as well, but if you still want to use the old tools, please download the documentation from a previous version.

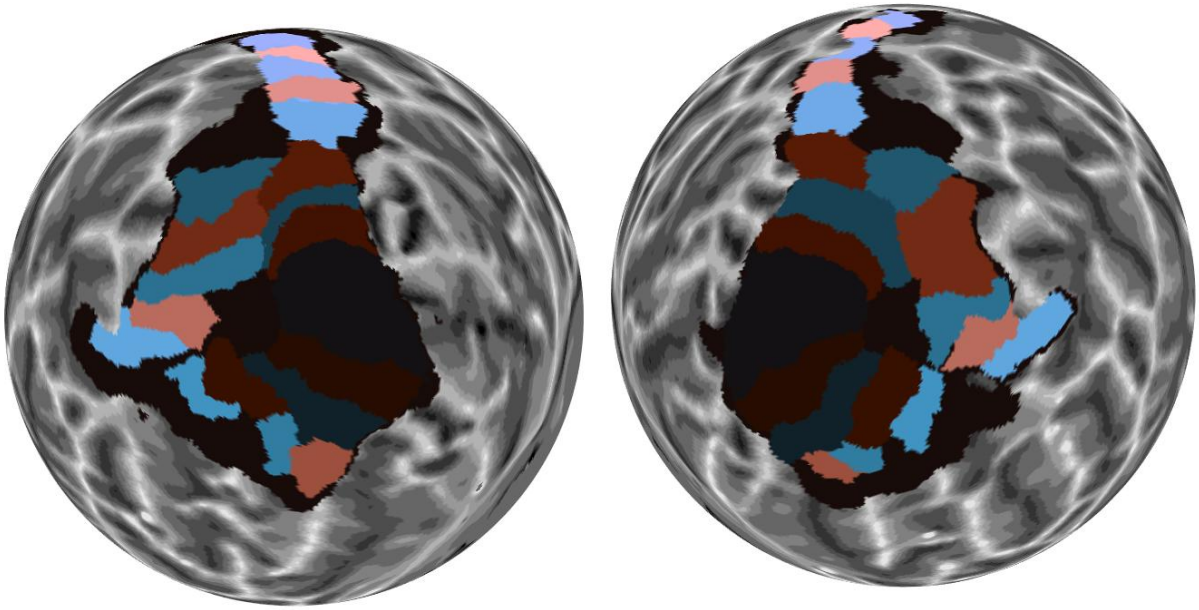
For manual delineation, we now recommend Fernanda Ribeiro's excellent web applet *SurfAnnotate*: https://webapps.neurodesk.org/surfannotate. This can load FreeSurfer files and GIIs for your surfaces and data overlays and saves out annotations in many formats (including a GII with annotations or FreeSurfer labels for individual ROIs). It also has superior drawing and filling algorithms.

Using template atlas maps and field signs

More importantly, in SamSrf we now have a new tool that can produce semi-automated delineations, and this tutorial explains how to use this. This requires template maps incorporating the field sign for each ROI to be warped back into native space using Template2NativeMaps
At the time of writing this, *only version 5 of the InfernoSerpents* template atlas contains field signs. You can download this from the SamSrf OSF repository[1]. This may look like this example to generate a template atlas map in the same native space as the empirical map *bi_pRF_Gaussian*[2]:

```
Template2NativeMap('bi_pRF_Gaussian', '../surf', 'Z:/InfSerpTemps_v5')
```

This atlas map contains the ROI labels with their field sign (see below what that means). Darker colours are regions with low numbers (like V1, V2, V3) and brighter colours are higher regions (like V3A, IPS0). The red and blue colour indicates that the field sign in that region is positive or negative, respectively.

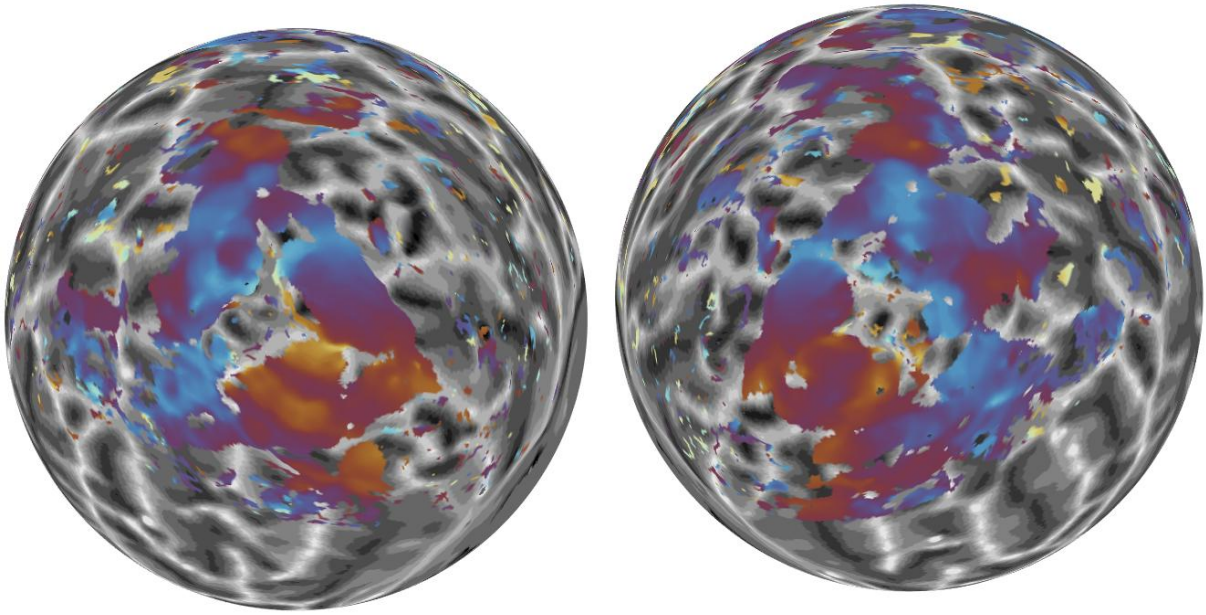


[1] Since OSF is shutting its doors to new projects, we will migrate future files elsewhere, but the files should persist there at least. Any potential future atlases will be uploaded in their new home however.

[2] You can also do this part in the SamSrfX GUI without any command line operations.

Our new delineation algorithm relies on these field sign maps. Please read Marty Sereno's original article on this (Sereno et al., 1995, *Science*, DOI: 10.1126/science.7754376). In short, the algorithm determines the angle between the gradient for polar angle and eccentricity maps, respectively, at each vertex on the cortical surface mesh. The sign of that angle is the field sign and basically reflects whether the visual field representation in a given brain region is an upside-down map like V1, or an upright map like V2. Using field signs to identify visual regions is not a new idea; however, this can often be complicated by the fact that field signs from neighbouring regions can merge or get distorted. In our approach, we therefore do not actually use the field sign, but the raw gradient angle.

An empirical map often looks something like the polar maps below (which are based on data where some surface smoothing was already applied when projecting volume images to the surface). A map like this would be fine for a practiced researcher to manually delineate visual brain regions, but things are more difficult for automatic procedures.



Creating field sign maps

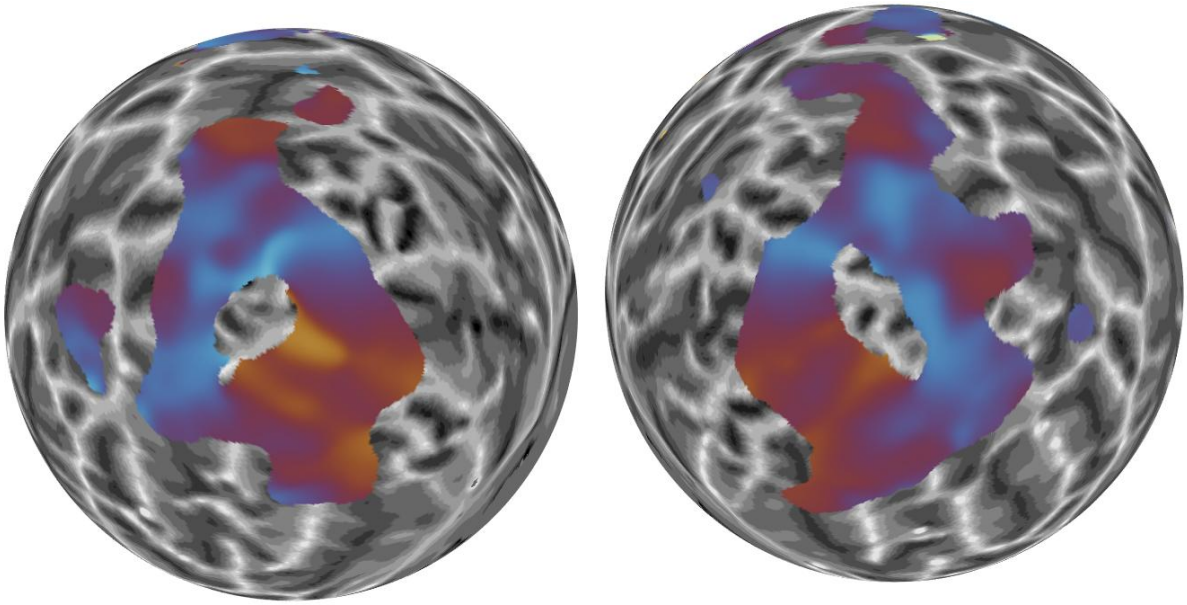
The gradient angle calculation would probably be quite noisy. To calculate decent gradient angle maps, you should therefore apply smoothing to this empirical map. Moreover, we want to filter out artifactual or noisy pRF estimates and restrict the map to a preset eccentricity range. The latter point is generally sensible to ensure consistency in the ROI definition and the resulting data analysis. To apply these operations, you can run the following functions in the command window or use the *Smoothing & Field Sign* menu option in the SamSrfX GUI[3].

```
load bi_pRF_Gaussian
Srf = samsrf_surfcalcs(Srf, 'occ', 0.5, [.5 8])
save('bi_pRF_Smooth', 'Srf', 'Model')
```

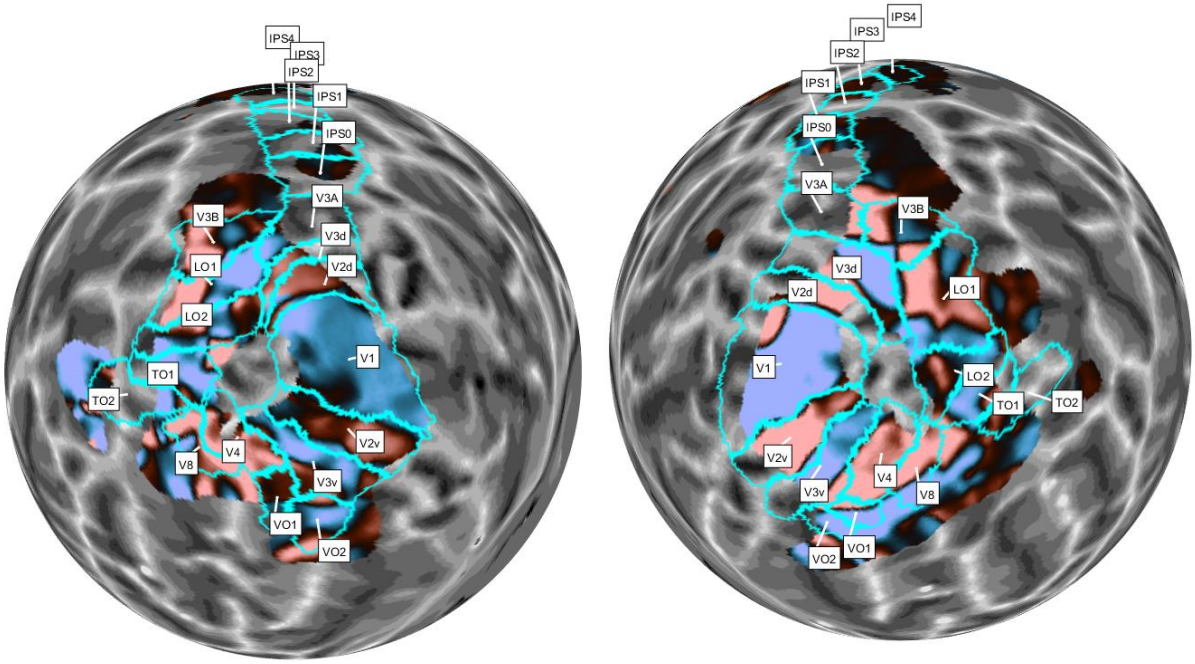
This uses mostly the default parameters, except that it limits the calculation to the occipital ROI occ, to pRFs with normalised $R^2 > 0.5$, and within the eccentricity range between 0.5 and 8 degrees. Obviously, your situation may differ. It then saves the smoothed maps, which also contain the gradient angle map, in *bi_pRF_Smooth*. Here are the resulting polar maps:

[3] But note that the surface calculations in the SamSrfX GUI have limited functionality.

Sam Schwarzkopf
Wednesday, 28 September 2026



And here are the gradient angle maps from the same maps. Warm colours are positive angles (upright maps) and cool colours are negative angles (upside-down maps). On top of this, I display the borders and the labels for the visual regions as determined by the template atlas. As you can see, these are roughly similar places. For some regions, especially earlier ones like V1, the atlas is already pretty accurate – but for other, especially higher regions, the atlas can be quite off.



Nevertheless, the correspondence between the template atlas and the empirical gradient angle map allows us to at least hazard some educated guesses as to where a particular region might be. The new delineation tool relies on this to suggest some locations. You will most likely need to correct some of these suggestions, and you also need to constrain the labels somehow. However, all of this can be done without manually drawing any borders between regions and it is completely reproducible because all the parameters of the algorithms can be saved and repeated at a later stage.

Sam Schwarzkopf
Wednesday, 28 September 2026

Using the fast delineation tool

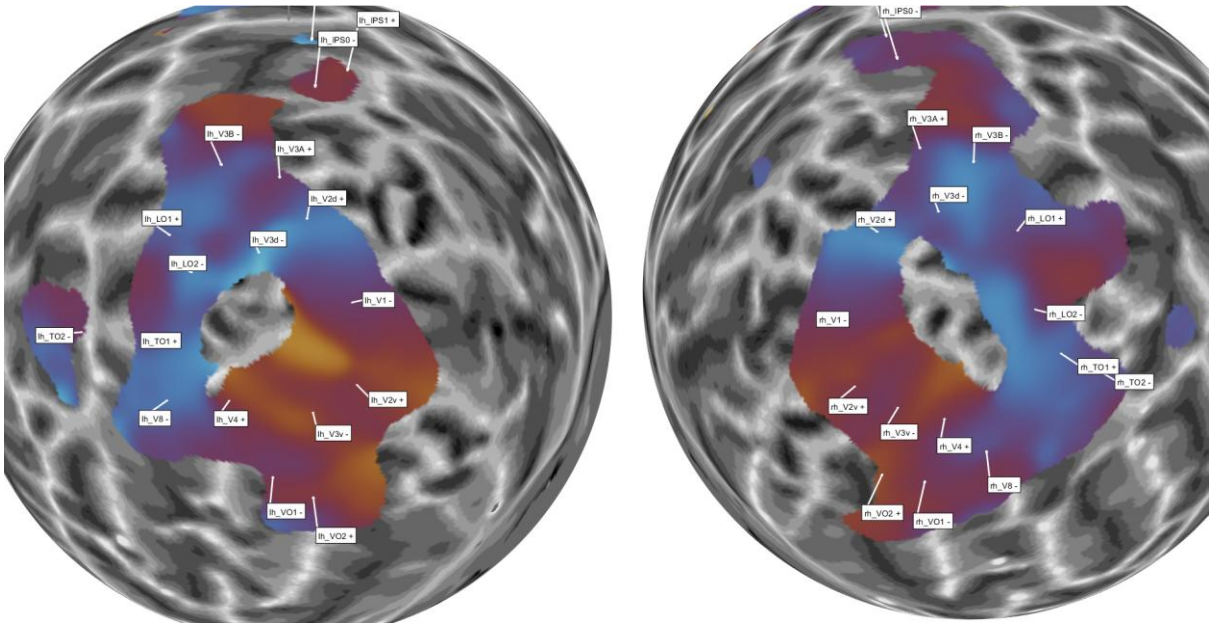
Here is how this works. You either run the tool in the command window or in the SamSrfX GUI:

```
FastDelineationTool
```

It will ask you to select the smoothed empirical map with the gradient angle, and the template atlas map with the ROIs and their field signs. In the command window, you can also directly specify these:

```
FastDelineationTool('bi_pRF_Smooth', 'bi_InfSerpTemps_v5')
```

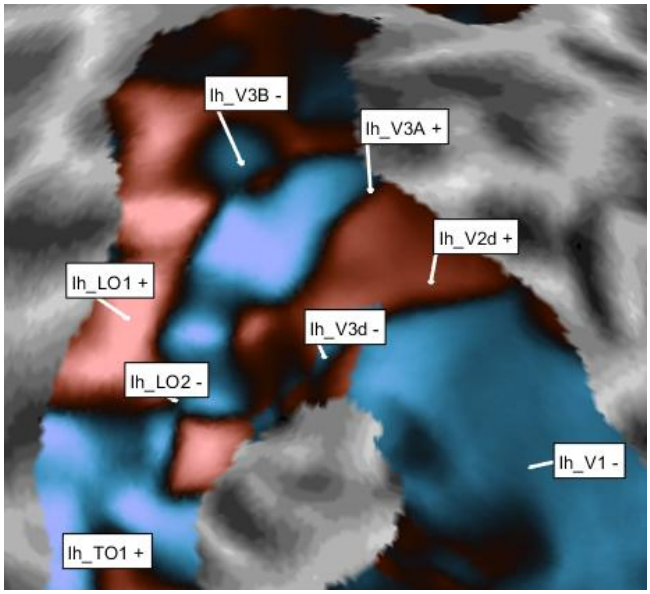
This will open the empirical polar map with the ROI suggestions from the template map. Note that these suggestions are based on where the nearest appropriate field sign might be – this can be quite wrong! While we may have refined these suggestions by the time you read this tutorial, I would not trust these suggestions to be accurate, especially for some higher regions. They are really meant to give you some guidance:



We decided to start this on the polar map because this is the one people are most familiar with. However, for assigning likely ROI locations it might make most sense to use the gradient angle map. You can select this view the *Maps* menu at the top of the window. You can also select various other maps like eccentricity, pRF size (sigma), or the actual field sign. There is also a ROI label map, but that one is currently empty.

Here, we continue with the gradient angle map. Our job is now to reassign any ROI suggestions that are not quite right to a more appropriate location. You do that via the menu *ROIs > Fix ROI Position* and then selecting a new location. This location needs to fulfil two criteria:

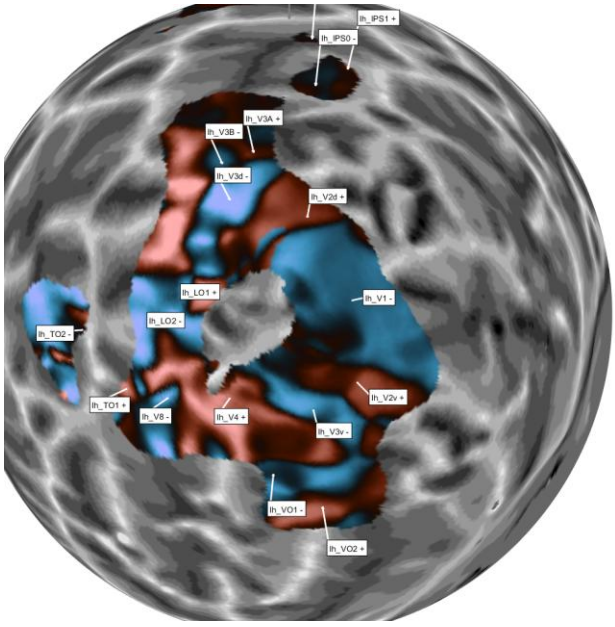
1. It needs to be in the correct hemisphere (this is obviously a given, if you do this analysis on surfaces from one hemisphere only – but our standard in *SamSrf* these days is to use bilateral surfaces unless there is a good reason to only look at one).
2. The field sign must be correct. You can tell the correct sign from the ROI label: e.g. V1 has the sign minus but V4 has the sign plus.

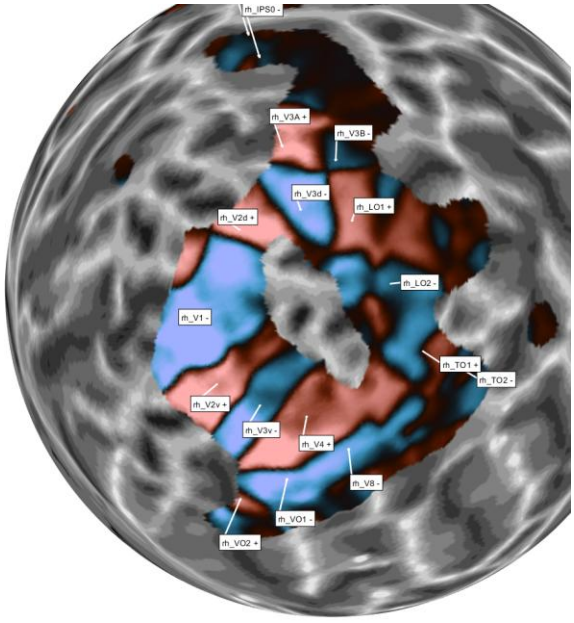


You can select as many vertices as you like until you found the location you want. In our example here, many of the ROIs already seem quite good. But as you can see in the image on the left, one clear exception is *lh_V3d*, which is in the wrong place: it is assigned to a spot that is actually in near the *lh_V1*-*lh_V2d* border. *lh_V3A* is also wrong as it is assigned to *lh_V3d*. The location of *lh_LO1* and *lh_LO2* are also off.

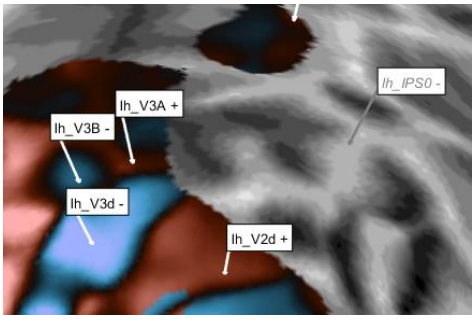
We therefore now reassign each of these one by one. Every time you selected a location for a ROI you're happy with, you can use menu to fix the next ROI location. But when you want to stop reassigning, you can use the *Rotation* menu to exit selection mode.

Below are the ROIs after I reassigned them, including the ones I changed in the right hemisphere. This obviously still takes some expert knowledge and decisions about where you think these ROIs should be. Part of this is practice with having seen many retinotopic maps, and having read the literature on these maps (see the second part of this tutorial about retinotopic organisation below). The delineation tool cannot take this job from you – even with a fully automatic procedure you would still need to know whether the delineation is in fact correct. Nevertheless, that caveat and warning aside, reassigning these ROIs is very quick and it is completely reproducible because the locations are saved. You can save your ROI seed selections and reload previous ones at any time via the *File* menu.



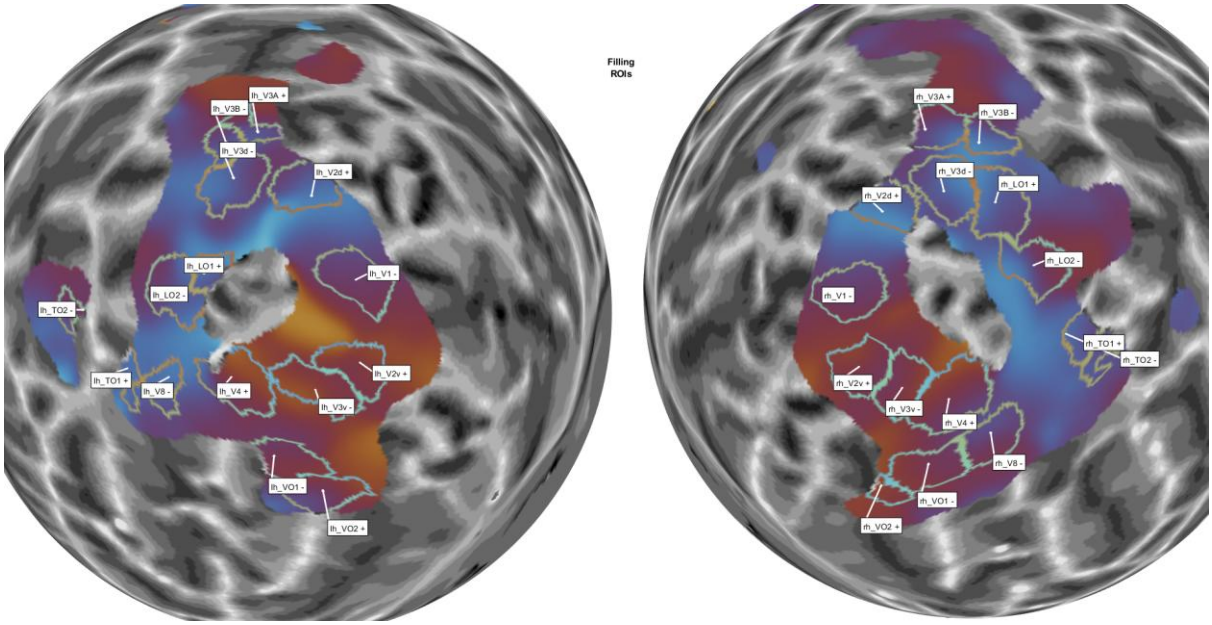


It is also possible to assign the ROI to an empty vertex (where the gradient angle is zero). This effectively takes the ROI out of the running so you can do that if you don't want that ROI to be delineated. Such a label would be greyed out, and the ROI will not be created. In our example, the IPS regions in both hemispheres are a bit ill-defined. I therefore selected an empty vertex for *lh_IPS0* (see image on the left).



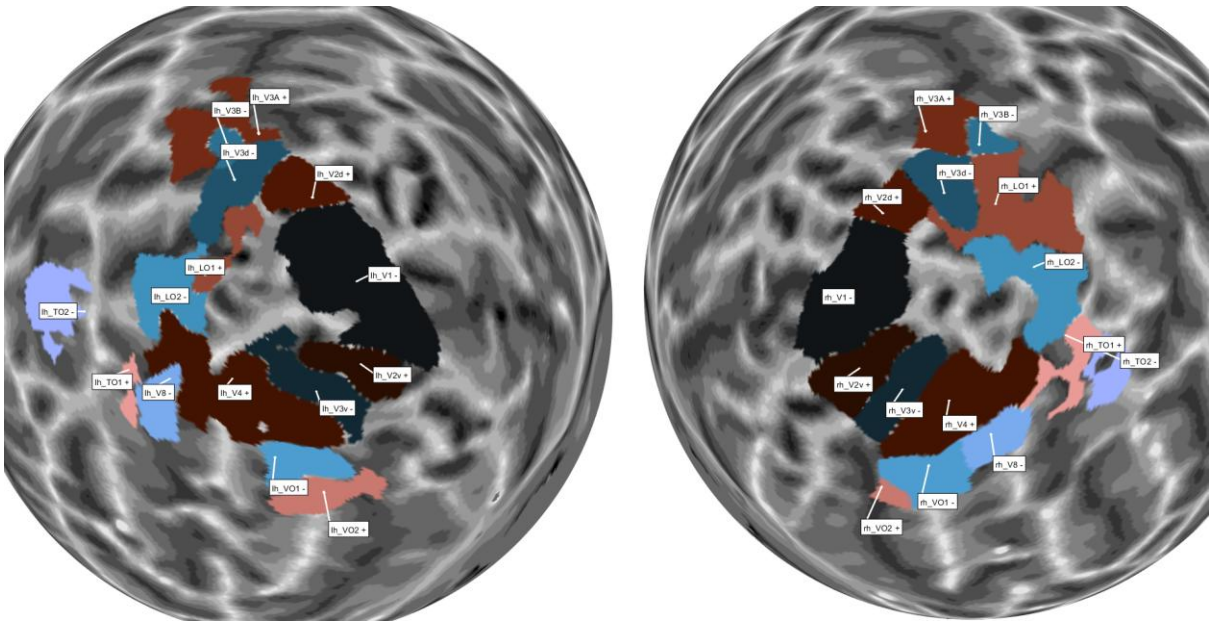
But doing this for several areas quickly turns tedious. You can speed this up by using the menu option *ROIs > Omit ROIs*. This allows you to select more than one ROI and all assign them to vertices with zero gradient angle (near the frontal pole).

In theory we could now use these assignments to determine the regions. You do this via the *Delineation > Fill ROIs* menu. This will first ask you to save your ROI selections (even if you may have already done that anyway) and then runs a flood fill operation to make the ROIs. This is how this might look half-way through filling:



Following this you end up with a ROI map, which you can also display via the *Maps* menu now. If you display any of the other maps (e.g. polar angle) you will see the newly labelled ROI borders. However, you will notice that the ROI labels have little gaps between them. We will deal with those later.

Another issue is that some regions clearly have been overfilled, such as *lh_V4, lh_V3A, rh_LO1*, and *rh_LO2*. The TO regions are also a bit dodgy, especially *rh_TO1*:

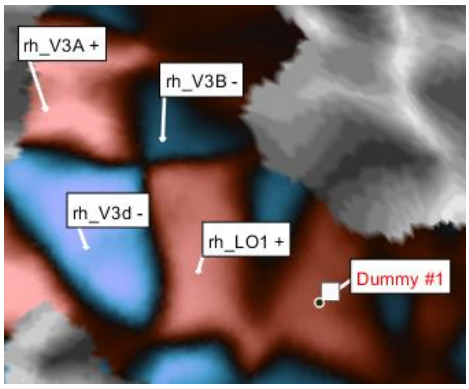


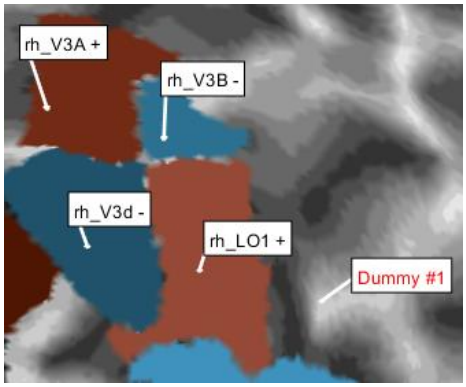
Sam Schwarzkopf
Wednesday, 28 September 2026

Adding dummies or new ROIs to contain overfilling

Let's start by blocking off that double area in rh_LO1. As you have seen, all the regions are filled simultaneously, starting from the vertex you selected for the ROI. When the flood filling from two ROIs meets, they stop each other. So, we can prevent some double region by selecting a dummy region using *ROIs > Add New Dummy*. This dummy region gets filled in as well but is then be disregarded. I have done in the example on the right. It will fill in rh_LO1 and the dummy at the same time.

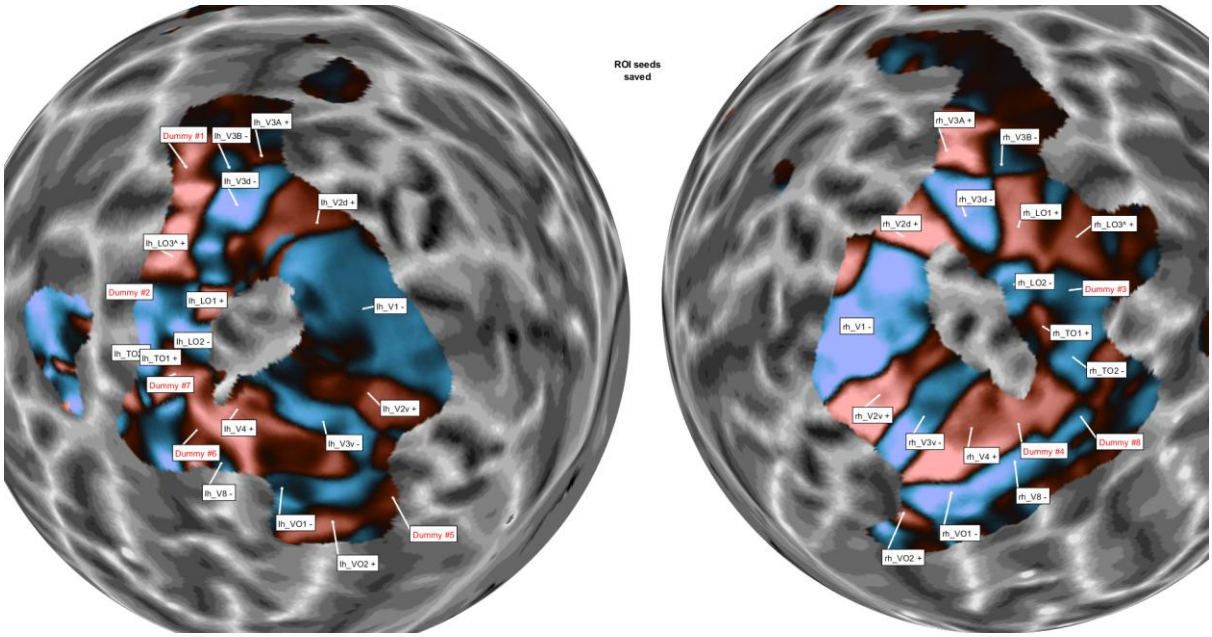
Note: Dummies fill regions with the selected field sign, whatever that might be – here it would be positive field signs.



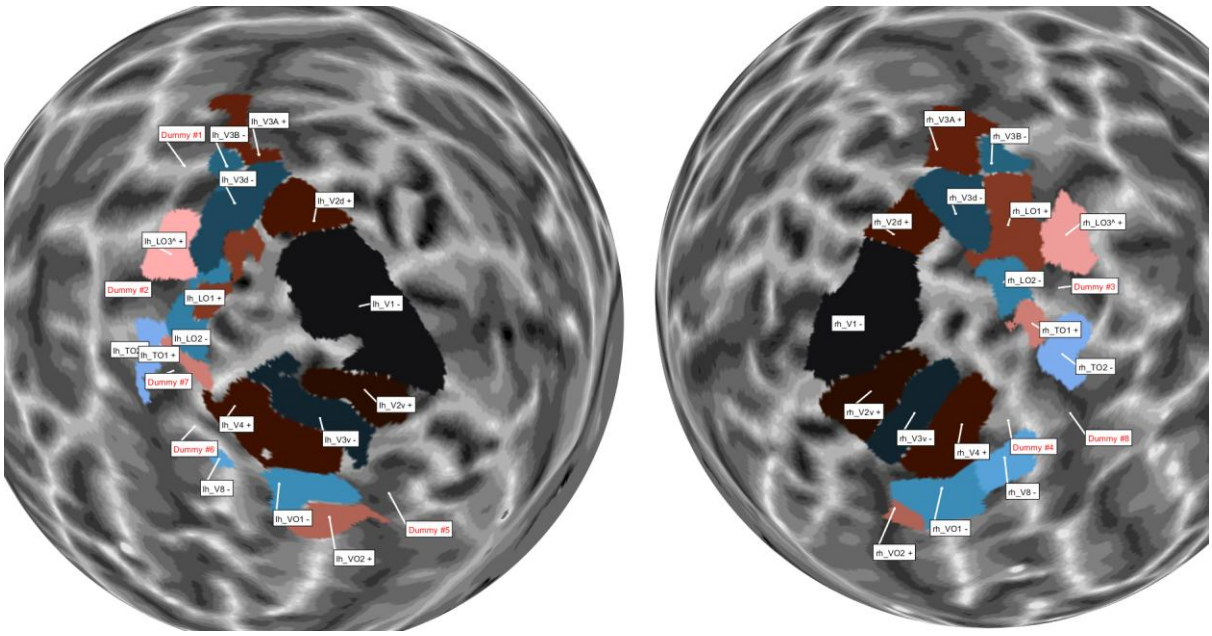


As you can see, after filling that the dummy was disregarded and rh_LO1 is now only the left portion. Obviously, this does not mean that region was meaningless. It was clearly a complete cluster of positive field sign. Judging by maps in the literature (esp. Sereno, Sood & Huang, 2022, *Front. Syst. Neurosci.* DOI: 10.3389/fnsys.2022.787737) this region is probably rh_LO3, but this region was not in our template atlas so there is no suggested label for it here. You could instead create a new ROI for this using *ROIs > Add New ROI*. This works just like adding a dummy except I have to also give it a name (including hemisphere prefix), so here: *"rh_LO3"*

You don't have to define the field sign for the new ROI because it will determine this from your selection. The ROI is suffixed with the hat ^ symbol (while writing this I learned this symbol is called a *circumflex*) to indicate that this is one you added manually rather than being based on the template map. We mainly need this so the program knows that you can remove it again if you decide to (see below). As you can see this ROI is now labelled *"rh_LO3^ +"*, but when we save the ROI labels in the end the hat symbol is not included in the filename. I have done this here and also in the left hemisphere where there is also evidently another LO3. Moreover, I have now moved some ROIs around (e.g. V8 and the TO regions in the left hemisphere) because they didn't look quite right. And I have blocked off some outlying clusters using dummy ROIs:

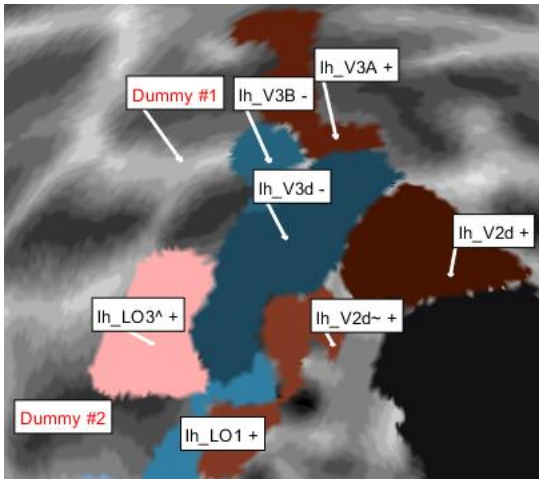


As I am now happy with these ROI selections, I can run *Delineation > Fill ROIs* again. Obviously, I had to make some judgement calls here so we won't pretend there isn't any subjectivity in this. The choice of where you put the seeds has a direct effect on how far the flood filling will reach so it affects where the region borders will end up. Usually, locations near a seed vertex will be filled in sooner, so you can probably predict where the filling will stop. If things still don't look right after filling in, you can adjust the locations further. But the point is, that all the seed locations are saved for posterity so others can later easily reconstruct how the regions were defined. If they disagree, they can adjust it.



Using duplicate ROIs to fix incomplete or erroneous fills

I am now mostly happy with how this looks but there are some flaws. Most notably, in the left hemisphere you see that lh_V2d does not seem complete. It does not reach down to the parafoveal representation as V1 or the other early regions do. What is worse, lh_LO1 has clearly filled too far. We could potentially fix this quite easily by simply moving the see for rh_V2d more centrally. However, another option you can use is adding a seed for that region using *ROIs > Duplicate ROI*. As you can see in the image on the right, I added such a duplicate in that erroneous bridgehead of lh_LO1. Now when I fill the ROIs, these both become lh_V2d, thus stopping lh_LO1 from encroaching.

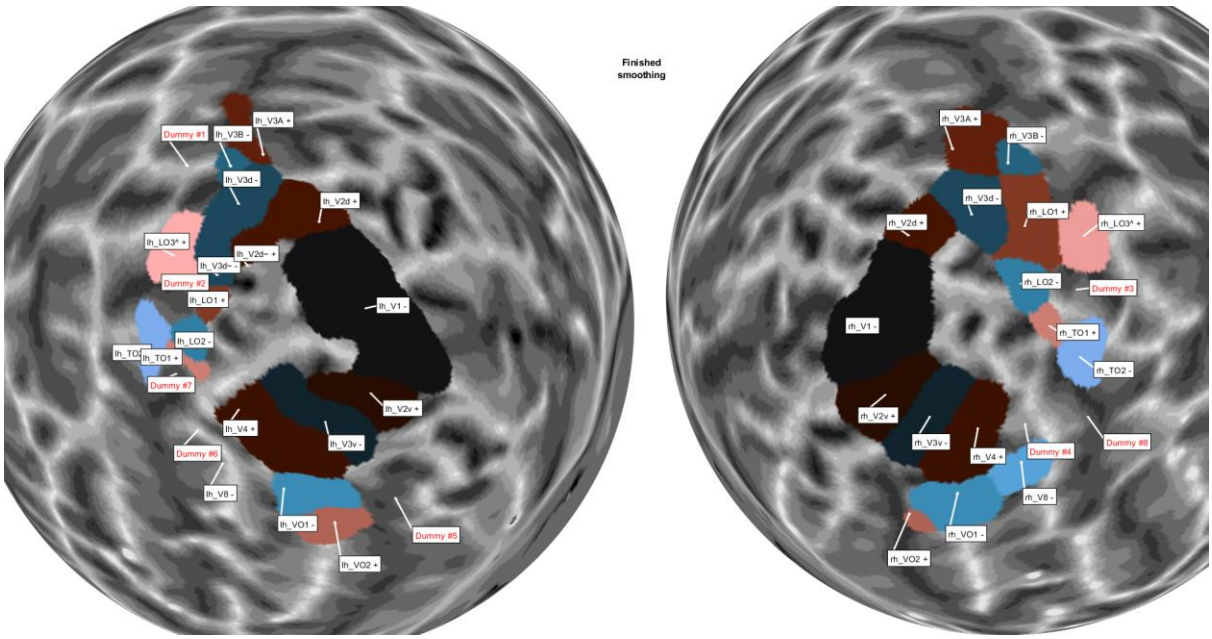


A similar issue exists for lh_V3d and lh_LO2, where the latter encroaches on the former's territory. We will fix this one using the same approach (not shown here).

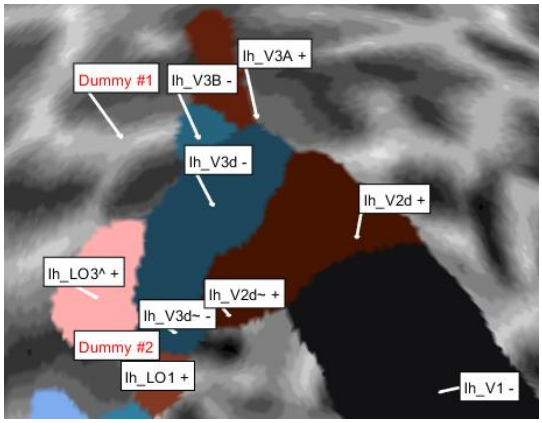
Sam Schwarzkopf
Wednesday, 28 September 2026

Smoothing and plugging gaps

If I'm happy with how the field sign clusters are labelled, I can now use *Delineation > Smooth ROIs.* This passes a simple nearest neighbour kernel over the ROI labels which will fill in the narrow gaps between the labels and also smooths out some of the jagged edges. Note that this operation runs over all the labelled vertices, so it can take some time depending on your computer (there is a progress bar in the MatLab command window).



But I'm still not fully satisfied. Specifically, V2d in both hemispheres look a little incomplete, especially in the left hemisphere. To fix that, we can use *Delineation > Plug Small Gaps* to fill a specific region in further. It asks you to select the region you want to fill. Note that this includes that region and all its duplicates you might have made. It will then perform *one* step of filling for that region. It fills in any vertices where there is gradient angle data, and that have not been filled in with another ROI label. The image on the left shows this after I did this for lh_V2d *three* times, and then I ran the smoothing again.



Note that plugging gaps no longer cares about the correct field sign! While it does not push into already labelled regions, and it won't fill empty vertices, it is designed to fill gaps without considering the field sign.

Saving ROI labels

Now we're finally ready to save our delineated region labels via *Delineation > Save ROI Labels*. This will first ask you save the ROI seed vertices you defined. These are stored in a JSON file, so it is easy to read. Basically this contains the seed locations of all the ROIs you defined, including new ROIs, duplicates, and dummies. So, by loading this file via *File > Load ROI Seeds* you can reload your previous work and rerun the filling at any time (and of course, you can also save your progress halfway through via the *File* menu).

Once the seeds are saved, the tool automatically saves the ROIs as FreeSurfer label files in a separate folder. If you delineated a bilateral surface file, the ROIs are combined across hemispheres. The folder also contains *Delineation.log*, which contains all the smoothing and gap plugging steps you performed.

Sam Schwarzkopf
Wednesday, 28 September 2026

Some final parameters

You may have noticed in the Delineation menu that there are some parameters you can define. These determine the behaviour of the flood filling operation. All these parameters are also saved in the ROI seeds file you save. This means you can transparently report how these parameters were set. You can repeat the filling exactly, and if you follow the *Delineation.log* you could also reproduce the exact smoothing and gap plugging steps. In my example and my various test uses of this tool, I simply stuck with the default and this worked well. But your mileage may vary. The filling parameters are as follows:

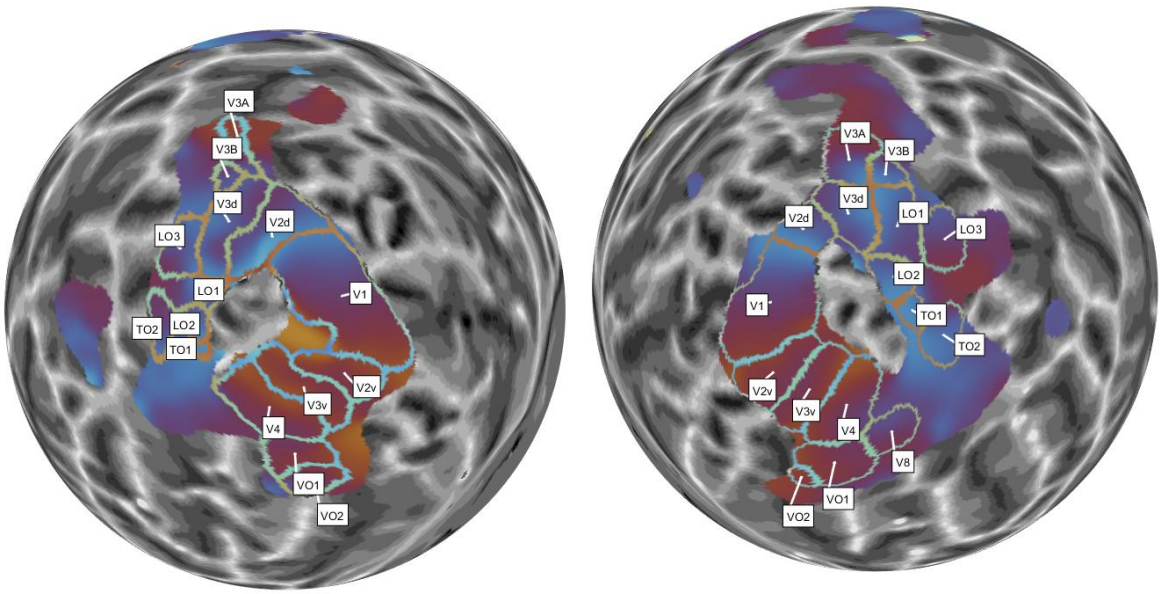
Max Cluster Size (Default = 300)
This defines the maximum number of vertices that will be filled in *if the fill has already reached some barriers*. This last part is quite important. The parameter does not mean that only ROIs up to 300 vertices will be created because that would be really small! It would not be able to fill a region like V1. Rather this is used in conjunction with the *Gap Tolerance* parameter. The idea is that a small region surrounded by other regions (like V3B) is not filled in further even if there is a small gap left.

Gap Tolerance (Default = 15)
The flood filling works by finding the border vertices surrounding the already filled vertices. Any vertices with the correct field sign and whose gradient angles are above threshold – and which haven't already been filled in by a different ROI – will be removed from the list. The *Gap Tolerance* parameter determines how many border vertices must remain to continue filling in the ROI, provided that the *Max Cluster Size* has not yet been reached. This can help constrain small ROIs.

Min Gradient Angle (Default = 0.1)
This is the threshold level of gradient angle that will be filled in. This incorporates the field sign of the ROI that is being filled. The higher this value, the less of a ROI will be filled in and the larger the gaps between labels will be. Larger gaps might not be a problem as you can still smooth the map.

Checking your ROIs

It is a good idea to inspect your labelling all went according to plan. You can now load the maps and ROI labels in the *DisplayMaps* tool and display the labels. This is how this looks for my example map:



Sam Schwarzkopf
Wednesday, 28 September 2026

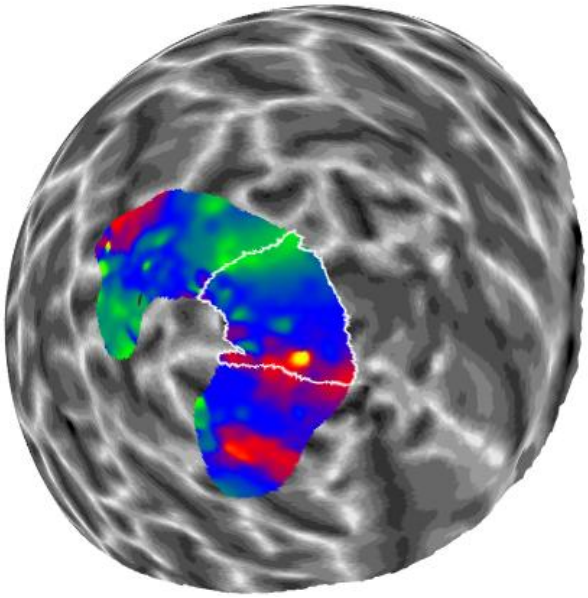
Delineation guide for visual areas

Delineating the earlier visual areas V1-V3 is usually quite straightforward, but even with them there are often tricky cases and open questions. Many higher areas are more confusing. Here I try to give some guidance as to how I have been delineating these cases. For a basic background you may also enjoy reading my simple (and now very old) retinotopy tutorial for phase encoded analysis:
https://doi.org/10.6084/m9.figshare.1513839.v1

More special cases may be added in future versions of this tutorial. Note that the images are created in *SamSrf*, using earlier delineation functionality. But this does not change the principles behind defining the areas. Please let me know if you have any questions, comments, or corrections about the advice given here:

V1

V1 contains a full hemifield map and is located fairly accurately within the calcarine sulcus. Thus it should span from a green stripe on the cuneus (dorsal bank of calcarine) through blue in the depth of the calcarine to a red stripe on the lingual gyrus (ventral bank)[4].



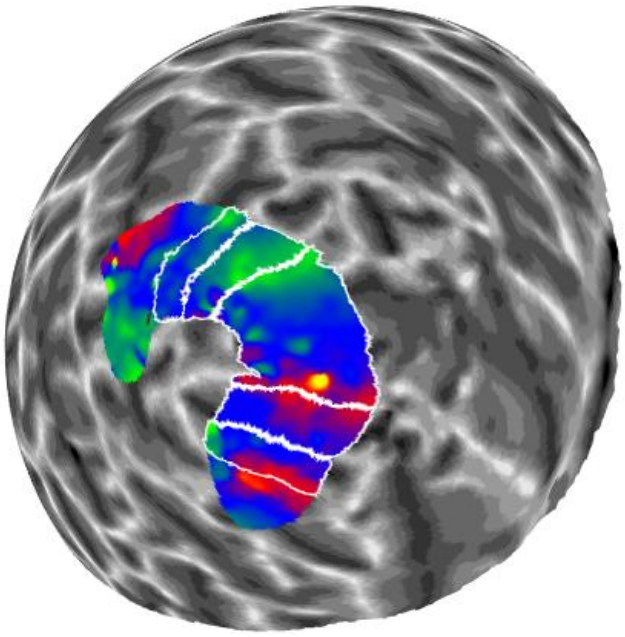
Note that the red/green stripes may not always be continuous. In this example (which I would say is actually a pretty good map) the red stripe on the ventral side is disrupted by blue near the occipital pole. Such interruptions are very common and likely artifacts of the surface sampling of the functional voxels. In worse cases the colour may even been green (which is much more worrying but fortunately fairly rare). You need to use your common sense a bit. Borders between regions don't just zigzag erratically around the brain. Changing the R^2 threshold may help remove rubbish data. Moreover, according to the theory that mechanical tension due to intracortical connectivity causes the cortical folding, the vertical meridians (red and green stripes) should coincide with the crest of the gyri. Especially in V1 this is typically the case and this is why anatomical predictions of retinotopic maps work so well. You can see this here somewhat although the correspondence isn't perfect.

[4] Note that these example maps use an old red-blue-green colour scheme that has been decommissioned. By default, SamSrf nowadays uses colour schemes that are supposedly better for people with colour vision deficiencies. However, we do not have the time to replace all these figures in the old tutorial. We apologise for any inconvenience caused.

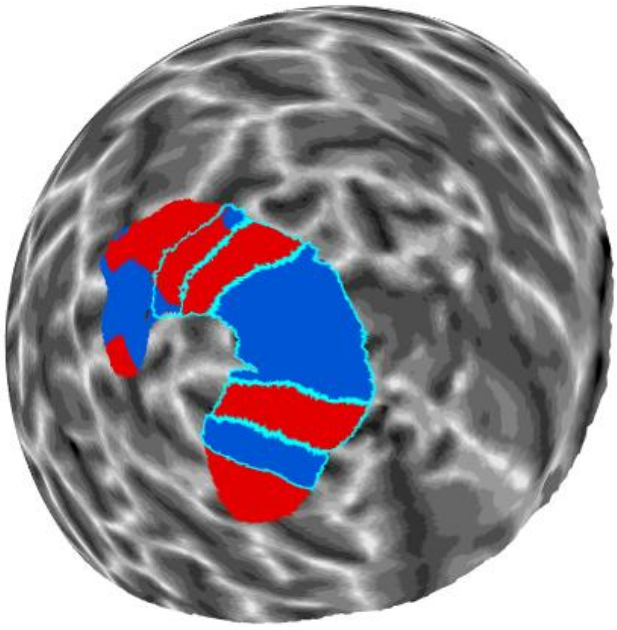
Sam Schwarzkopf
Wednesday, 28 September 2026

V2 and V3

V2 and V3 are both segregated into two quadrant field maps, one in ventral cortex and one on in dorsal cortex. So V2d runs from the green stripe of the V1 border to the middle of the blue stripe. V3d then runs from that blue stripe to the next green stripe. Conversely, V2v runs from the red stripe of the V1 border to the blue stripe, and V3v runs from that blue strip to the next red stripe.



In the example above, the ventral borders are pretty clear. However, the dorsal borders are trickier. The V2d-V3d border is disrupted by a washed out bluish green and the outer border of V3d is interrupted by a large blue gap. The dorsal borders between V2d and V3d I drew manually but I used the field sign blobs on either end as a guide as to where these areas should probably be:

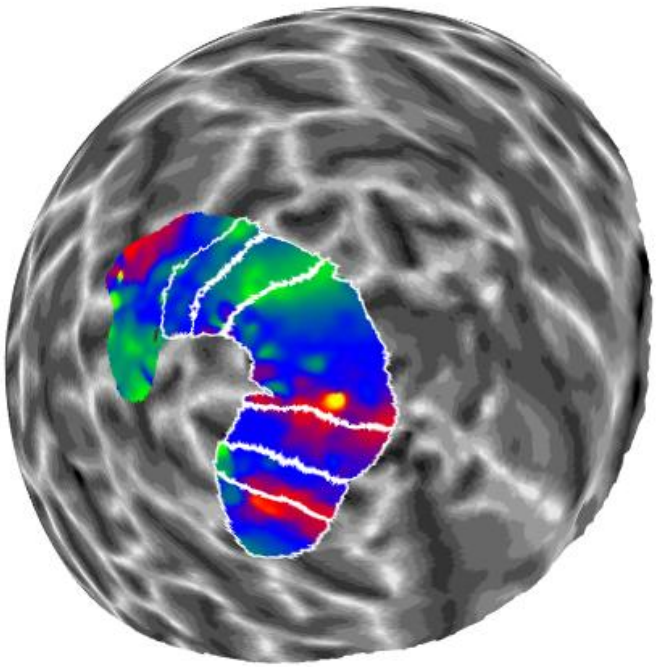


I simply continued the trajectory of the little blue blob on the peripheral edge to tell me the likely borders of V3d to gradually converge towards the foveal confluence. For the outer border of V3d I also used the polar map to connect between those green islands. **Critically, you want to go for the peak of the (in this case, green) colour.** This all seems fairly reasonable to me but as you can see cases like this requires a judgement call.

It's hard to know what the correct answer is here. This scenario, a washed out green regions where V2d and V3d should be, is surprisingly common. Again the field sign and eccentricity maps will help a little here because it tells you how far the occipital fovea extends (which tells you how far V3d should go) and it may also segregate the upside-down map of V3d from the normal map in V2d. It is also important to use the presence of V3A as a guide. In these maps we clearly have a full hemifield from green to blue to red indicating V3A (see below). This tells you that anything posterior to V3A is presumably V3d.

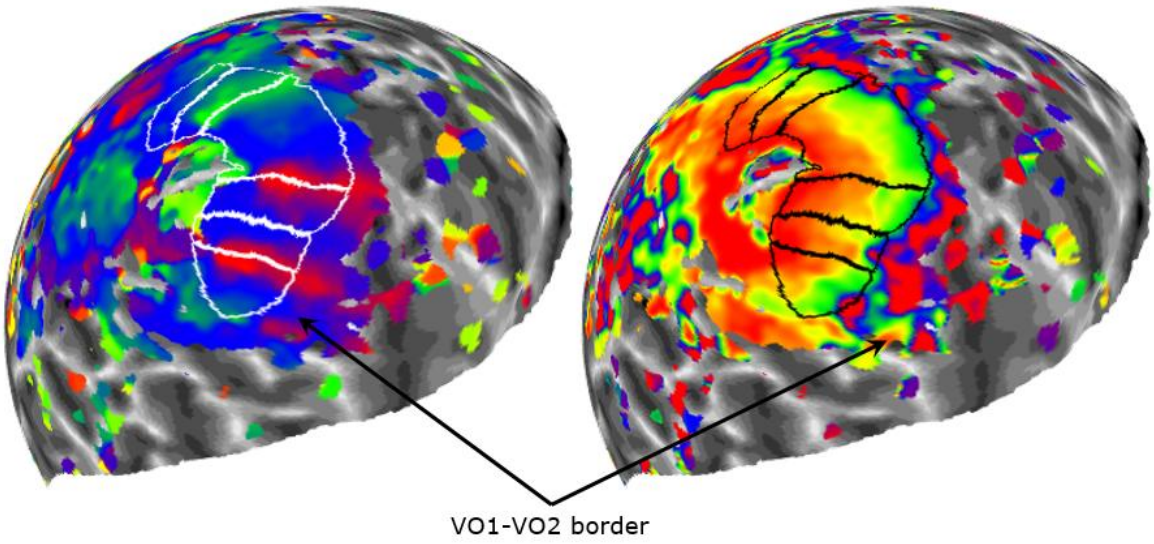
V4

Some people call this hV4 to distinguish it from monkey V4, which appears to be organised differently. Whatever you call it, based on my own experience at least, human V4 always contains a hemifield map of the contralateral visual field. So it should run from the red stripe at the outer border of V3v to a green stripe. This green stripe is often disrupted by blue blobs – but in my experience there is almost always at least a bit of green, especially towards the more peripheral parts of V4.



In fact, this green peripheral bit indicates the peripheral border of V4. This is a polar angle reversal around a lower vertical meridian (just like the border between V1 and V2d). At this point the green outer border of V4 (that is, the one that is parallel to the border with V3v) makes a 90 degree turn towards V3v. Anything beyond this green reversal along what would be more peripheral V4, if V4 behaved like V1-V3, are the VO areas. They are VO1 and VO2 and possibly more. VO1 and VO2 contain a full hemifield as well and they share a fovea that is *separate from the occipital fovea*.

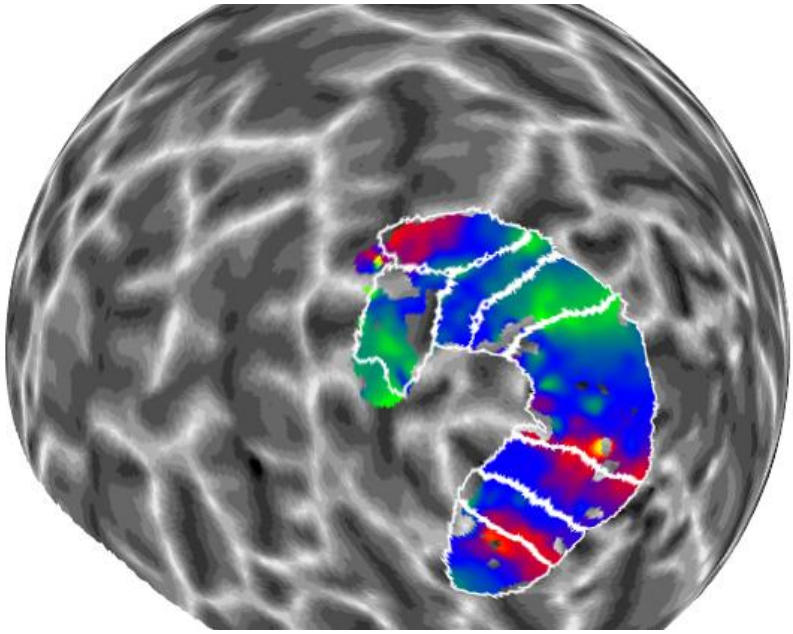
All of this should be visible in the maps below. This is from the same participant but a different analysis and using different thresholds. I didn't draw labels for VO1 and VO2 (because my labels are based on the smooth field sign map where I didn't see these) but I indicated what seems to be the border by arrows. This location corresponds to an upper vertical meridian (red polar angle) and a foveal representation (red in eccentricity map). In the best-case scenario, you should however be able to distinguish them via the field sign also: VO1 should be blue (mirror map) and VO2 red (normal map).



If you paid attention, this means that V4 is heavily biased towards the central visual field because it either doesn't represent the periphery at all or in an extremely compressed way. This arrangement is similar to the relationship between V3d and V6 – but since I have no maps that show that I won't show you any examples of that for now…

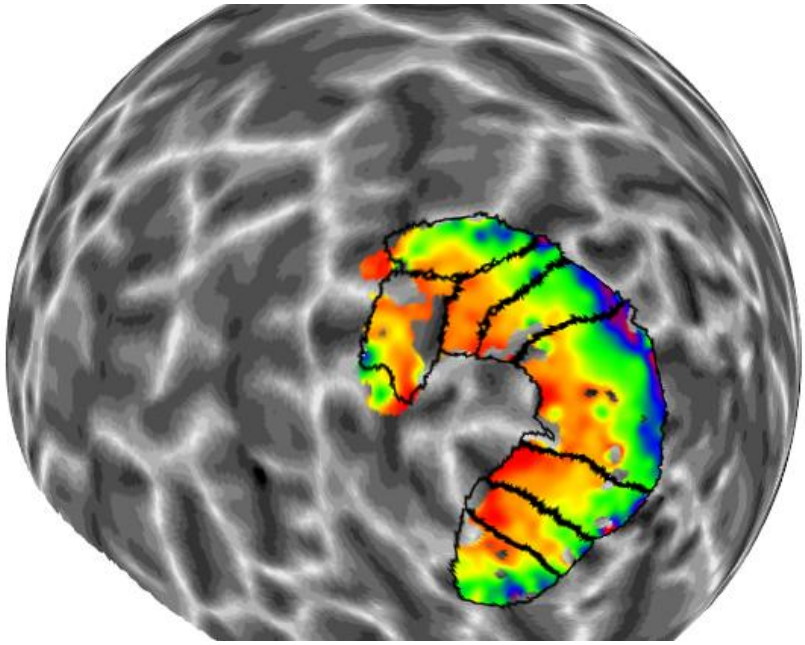
V3A and V3B

V3A is probably the most easily recognisable area after V1. It contains a full hemifield and is dorsal and anterior to V3d. Even in awful maps you usually see a hint of the red border characterising the anterior extent of V3A. Next to V3A, on the side pointing towards the occipital pole, you will find V3B, a rather peculiar area that nobody seems to quite agree on.



In really good maps, you will see that the outer red border of V3A suddenly takes a sharp 90 degree turn veering away from the occipital pole in an anterior-ish direction. This feature indicates one border of V3B. The other border should be a green reversal.

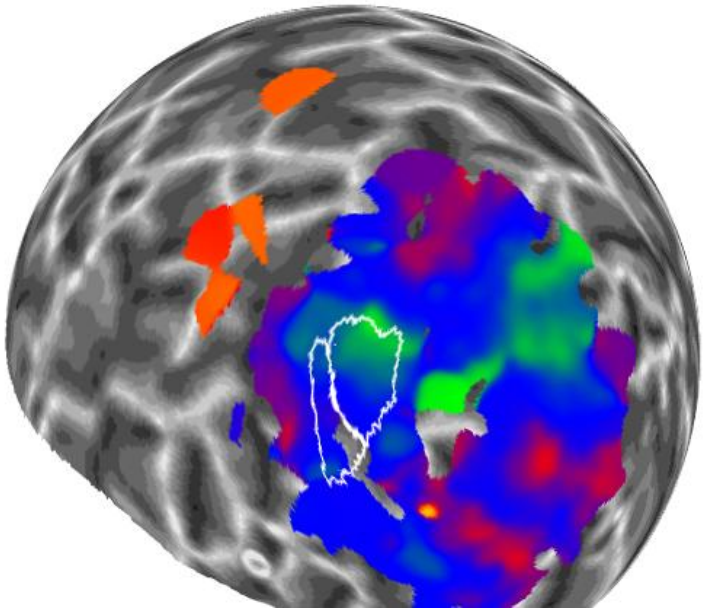
Critically, V3A and V3B share a common fovea which is usually different from the occipital fovea, even though the two foveas are usually attached. In the example below it looks as though it is the occipital fovea. In other brains, the distinction is more obvious because the eccentricity gradient makes a sharp turn out there.



LO areas

LO is composed of (at least) LO1 and LO2. A poster (possibly a publication too by this point?) by the York group claimed that there are also LO3-LO6, which are located more laterally in the region that many traditional localiser types call LO and/or OFA (in fact, they suggested that LO3-LO6 *are* OFA). You often need a lot of imagination to see the LOs and I can't honestly say I have made out any maps beyond LO1 and LO2.

LO1 and LO2 are supposed to contain full hemifield maps. So the border between them should be a red stripe (upper vertical meridian) running towards the occipital fovea, which both these regions share. On the other side both LO1 and LO2, should be bounded by a green stripe (lower vertical meridian). I have rarely seen this arrangement. Much more likely it looks as in the example below, where we have mainly blue (horizontal meridian and at best a tiny purple-red spot. This is where I drew the border between LO1 and LO2.

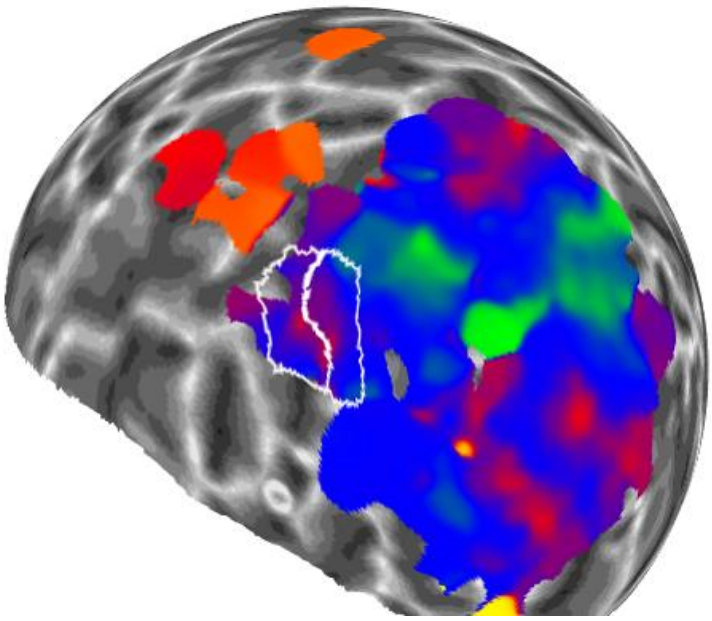


You'll notice that in these maps I dropped the threshold to show a lot more. The strictly thresholded and eccentricity-bounded maps I showed before wouldn't really show anything here. I also removed the earlier areas here to avoid confusion. The blue outer border of LO2 should really be green (and there was a hint of green here) because LO2 is supposed to be a full hemifield.

MT+ complex (TO areas)

This brings us to the MT+ complex. People of the original London school of thought may choose to call this V5, in honour of Semir Zeki's naming of that area (and so you also often see the rather awkward V5/MT in the literature). The important thing to keep in mind here is that MT is only a small area that is part of the motion sensitive complex in the medial temporal cortex.

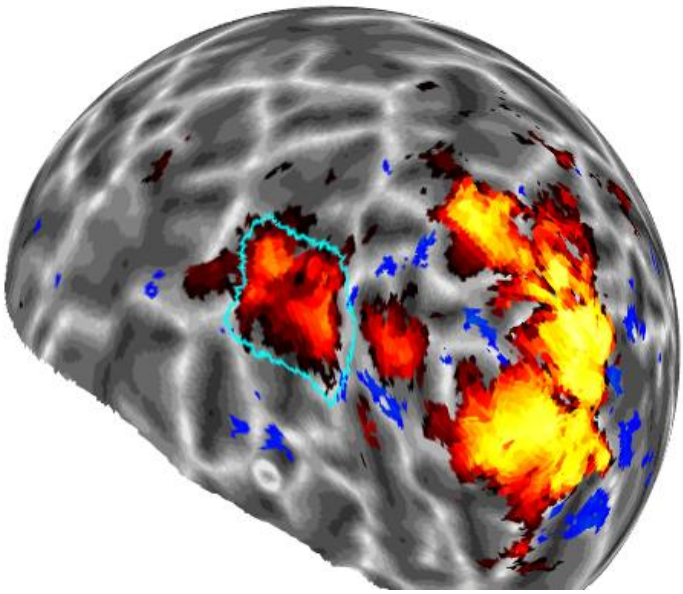
According to the Wandellian nomenclature, there are actually at least two retinotopic regions here, anterior to LO2. They are once again full hemifield representations although you may not see the lower vertical meridians (green stripes). You do tend to see the upper vertical meridian (red stripe) where the field sign for those areas reverses. Going from posterior to anterior they named these two areas TO1 and TO2.



It has also been shown that TO1 is the actual classical area MT (or hMT, to distinguish it from the monkey literature). TO2 is probably area MST which supposedly has somewhat different response properties. These two areas also have a separate foveal representation although that can be difficult to make out. There may be more areas that are part of this cluster. If you ask Marty Sereno, he will tell you that there are lots of small areas in this part of cortex (similar to the reports about the LO areas mentioned earlier).

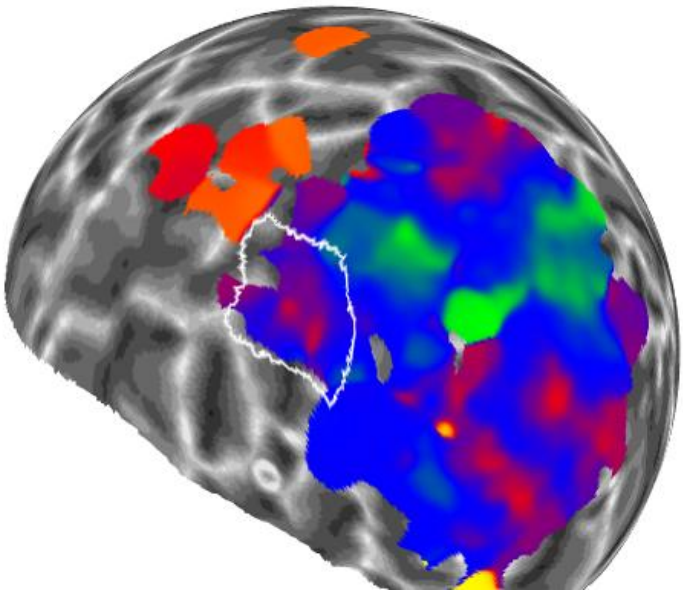
Critically, since this is the motion-sensitive MT+ complex, we should measure stronger responses to moving than static stimuli here. This is how original studies identified this region. They would therefore have been agnostic about any retinotopic structure that can distinguish sub-regions.

Below, I show the activation maps for a GLM contract comparing alternating blocks of radially moving dots with blocks of static dots – a standard MT+ localiser scan. You can clearly make out a large activation all over the early areas V1-V3 and V3A. But there is also a clearly distinct cluster in medial temporal cortex. (Note that these maps are based on data that had been smoothed in the volume prior to the GLM analysis, hence the different granularity than in the pRF maps):



I drew a rough region of interest around this medial temporal activation. There is a little cluster in this particular map between that and the large one on the right. This is inside the LO complex and could correspond to parts of LO1 (which is probably what had been called KO, although that may also overlap with V3B). Anyway, we will not include this small patch but just the larger one. I now overlay this delineation on the retinotopic map:

Sam Schwarzkopf
Wednesday, 28 September 2026



The overlap is striking. This motion sensitive cluster quite clearly corresponds to the location of TO1 and TO2. Of course, things may not always work out this way and there are several factors that could screw up this relationship (such as bad coregistration or simply bad maps). But as far as I managed to see these TO areas and had a localiser at hand, the correspondence was usually very close. So in theory, if you get good pRF maps in these regions you should be able to map the MT+ complex without a motion localiser. However, a localiser may nevertheless be useful because it is a lot easier to identify when the retinotopic maps are poor.

Motion sensitivity may also be of interest elsewhere by the way. One thing I noticed is that V4 shows little response to the motion localiser, but the adjacent VO regions may do.

Sam Schwarzkopf
Wednesday, 28 September 2026

Reading List

Here is a list of publications that you should look at for getting a better understanding of how these maps are organised. They also show you how large the individual differences can be and how there still remains disagreement about some of the regions:

Seminal study explaining visual area borders and field sign reversals

Sereno, M.I., Dale, A.M., Reppas, J.B., Kwong, K.K., Belliveau, J.W., Brady, T.J., Rosen, B.R., Tootell, R.B., 1995. Borders of multiple visual areas in humans revealed by functional magnetic resonance imaging. Science 268, 889–893.

Good general overview over the visual field maps

Wandell, B.A., Dumoulin, S.O., Brewer, A.A., 2007. Visual field maps in human cortex. Neuron 56, 366–383. https://doi.org/10.1016/j.neuron.2007.10.012

Comprehensive atlas of all manner of regions (not just visual!)

Sereno, M.I., Sood, M.R., Huang, R.-S. (2022): Topological Maps and Brain Computations From Low to High. Frontiers in Systems Neuroscience 16. https://www.frontiersin.org/articles/10.3389/fnsys.2022.787737.

About the foveal confluence of the early regions

Schira, M.M., Tyler, C.W., Breakspear, M., Spehar, B., 2009. The foveal confluence in human visual cortex. J. Neurosci. Off. J. Soc. Neurosci. 29, 9050–9058. https://doi.org/10.1523/JNEUROSCI.1760-09.2009

About the clusters with separate foveal confluences

Wandell, B.A., Brewer, A.A., Dougherty, R.F., 2005. Visual field map clusters in human cortex. Philos. Trans. R. Soc. Lond. B. Biol. Sci. 360, 693–707. https://doi.org/10.1098/rstb.2005.1628

About V4 organisation and the VO regions

Winawer, J., Witthoft, N., 2015. Human V4 and ventral occipital retinotopic maps. Vis. Neurosci. 32, E020. https://doi.org/10.1017/S0952523815000176

About the TO regions and also the LO regions

Amano, K., Wandell, B.A., Dumoulin, S.O., 2009. Visual field maps, population receptive field sizes, and visual field coverage in the human MT+ complex. J. Neurophysiol. 102, 2704–2718. https://doi.org/10.1152/jn.00102.2009

Original study on the LO regions

Larsson, J., Heeger, D.J., 2006. Two retinotopic visual areas in human lateral occipital cortex. J. Neurosci. Off. J. Soc. Neurosci. 26, 13128–13142. https://doi.org/10.1523/JNEUROSCI.1657-06.2006

TMS study looking at specialisation of the LO regions

Silson, E.H., McKeefry, D.J., Rodgers, J., Gouws, A.D., Hymers, M., Morland, A.B., 2013. Specialized and independent processing of orientation and shape in visual field maps LO1 and LO2. Nat. Neurosci. 16, 267–269. https://doi.org/10.1038/nn.3327